

Yifan (Frank) Wang

frankwang3744@gmail.com | (212) 518-6933 | Milwaukee, Wisconsin
https://linkedin.com/in/yifanwang5 | https://yifanwang3744.github.io

EXPERIENCE

Elutions

Milwaukee, Wisconsin

Software Developer | C++ · C# · JavaScript · .Net · Next.js · React · PostgreSQL

May 2023 - Now

- Led development and support of **ControlMaestro**, a large-scale distributed **SCADA** software deployed worldwide, providing real-time data monitoring, control, and alarming for industrial systems; designed and developed **50+ new features**, resolved **200+ tickets** across backend and frontend
 - Desktop (C++)**: Created 30+ product features for core services of desktop application, including: an Alarm Inhibition System enabling custom alarm filtering; an Application Upgrade Tool that simplifies software upgrade process and reduces downtime; optimized multi-threaded process synchronization with critical section objects; and a Cross Reference Tool for tracking tag usage and enabling real-time communication with **SQL Server** databases
 - Web (React/C#/.Net)**: Upgraded the full-stack React web application, improving responsiveness and stability for real-time visualization while delivering performance and feature enhancements: fixed memory leaks by optimizing React state management and refactoring cleanup logic, achieving a **20% memory heap usage decrease** in Memlab tests; developed RESTful APIs and middleware services for BFF services; enabled full Unicode support through ANSI-Unicode conversion
 - Driver (OPCUA)**: Developed an **auto-reconnection mechanism** to improve connection stability, and an optimized server scan method with caching and batching services calls which **reduced scan time by 75%**
 - DevOps (CI/CD)**: Built and maintained CI/CD pipelines with **Jenkins** and **GitHub Actions**, streamlining releases and improving project planning and delivery tracking
- Led **0→1 development** of **Chromium** - a web-based SCADA application, designing a **scalable microservices architecture**, selecting appropriate technology stack, collaborating with cross-functional teams to drive product delivery
 - Defined and built four microservices and data streaming pipeline, including a **NATS broker** service, a **YARP gateway** service, a tag service using **Redis** as a high-speed live data layer and **PostgreSQL** for data storage, and frontend using **Next.js/React**
- Implemented and optimized an unsupervised data importer program in **Pascal** to collect binary data and convert to readable format, automated tasks with SQL stored procedures and scheduled tasks, contributing to securing a contract renewal with client

EDUCATION

Columbia University

New York, NY

M.S. in Electrical Engineering

Sep 2021 - Feb 2023

Coursework: Analysis of Algorithms, Applied Machine Learning, Introduction to Databases, Computer Systems, Cloud Computing

Sichuan University

Chengdu, China

B.S. in Electrical Engineering

Sep 2017 - Jun 2021

TECHNOLOGIES

Languages C/C++, C#, Python, Java, JavaScript, TypeScript, SQL, Pascal

Frameworks React, Next.js, .NET, Node.js, FastAPI, Django, Flask, Express, Redux, Angular, YARP, Spring Boot

Infrastructure AWS, GCP, Docker, Jenkins, Git, PostgreSQL, Redis, MongoDB, RabbitMQ, Hibernate, Spark, Hadoop, Unix/Linux

PROJECTS

Personal Project: HR Management System Microservices | Java, Spring, Eureka, Hibernate, MongoDB, Swagger, RabbitMQ

- Developed backend microservices for a HR management and employee onboarding application, created RESTful APIs with Spring MVC, utilized Eureka and RabbitMQ for service registration and communication
- Secured the application with an authentication service and JWT, implemented a email service for SMS, managed data pipeline with Hibernate, MongoDB, and Spring Data JPA

Columbia University: AWS Bookstore Application | Python, FastAPI, AWS (S3, Lambda, API Gateway, RDS), React, OpenAPI

- Constructed four microservices and RESTful APIs to implement an online bookstore application with FastAPI, implemented secured login and email notification utilizing OAuth2 and Amazon SNS, and managed data with AWS RDS
- Created a responsive UI/UX with React, encapsulated application with API Gateway and CloudFront, and deployed on AWS

Research Project: Optimized Localization Method for ORB-SLAM2 | C++, Computer Vision, SLAM

- Proposed and implemented an improved localization method based on image pyramids theory for ORB-SLAM2, a real-time SLAM system that computes camera trajectory, resulting in a 5% increase in localization accuracy